

ENDG 319

Probability, Statistics and Machine Learning

Lecture 6: Conditional Probability and Independence

Multiplication rule • Total probability • Bayes' rule

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Learning outcomes for Lecture 6

READ THE GIVEN INFORMATION BEFORE CHOOSING A RULE

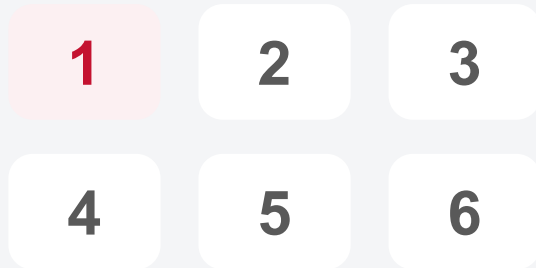
- 1 Condition** Calculate a probability after restricting attention to a given event.
- 2 Assess independence** Decide whether learning about one event changes another probability.
- 3 Multiply** Use conditional probabilities to find the probability of a complete path.
- 4 Combine and reverse** Use a partition for total probability and Bayes' rule.

Core question: “Among which outcomes are we now calculating the probability?”

Conditioning changes which outcomes we consider

ONE FAIR DIE: $A = \{1\}$, $B = \{1, 3, 5\}$, $C = \{2, 4, 6\}$

No extra information



$$P(A) = 1/6$$

Given B: the result is
odd



$$P(A | B) = 1/3$$

Given C: the result is
even

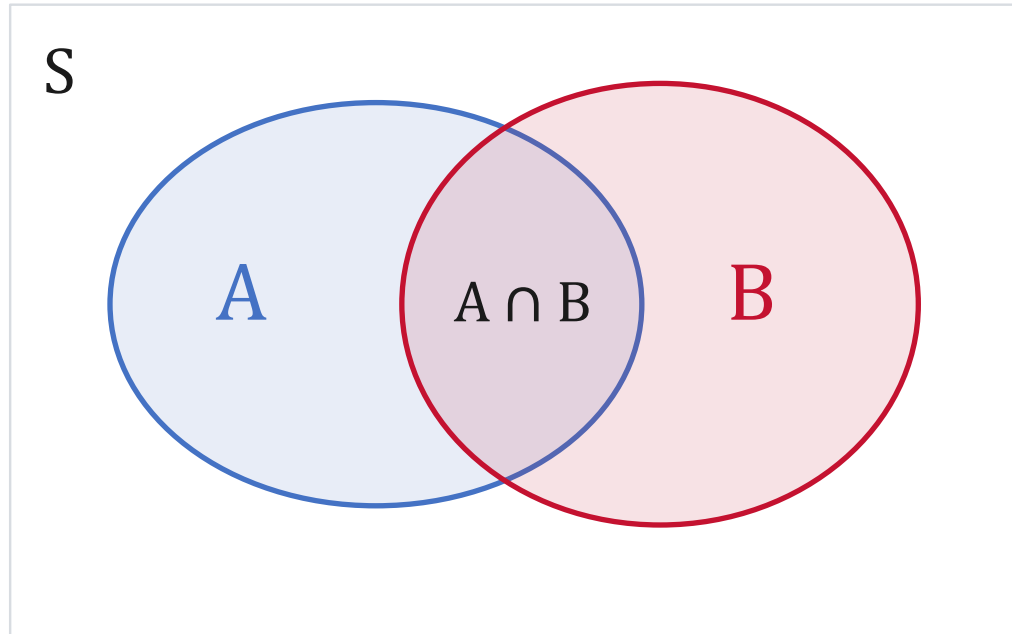


$$P(A | C) = 0$$

The outcome of interest is still “1”; the information changes the reference set.

Conditional probability: focus on the given event

A PROBABILITY WITHIN B, PROVIDED $P(B) > 0$



Schematic: areas do not encode numerical probabilities.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Numerator: outcomes satisfying both A and B.

Denominator: the probability of the given event B.

Restrict to B, keep the part of A inside B, and renormalize by $P(B)$.

A | B and B | A ask different questions

CHECKPOINT USING THE SAME DIE EXAMPLE

$P(A | B)$

Among odd results {1, 3, 5},
how often is the result 1?

$P(B | A)$

Given that the result is 1,
is it odd?

$$P(A | B) = 1/3$$

$$P(B | A) = 1$$

The intersection is the same, but the denominators $P(B)$ and $P(A)$ differ.

Independence: information does not change probability

TWO EVENTS A AND B

A and B are independent when learning whether one occurs does not change the probability of the other.

When $P(B) > 0$

$$P(A \mid B) = P(A)$$

When $P(A) > 0$

$$P(B \mid A) = P(B)$$

Equivalent product test: $P(A \cap B) = P(A)P(B)$

The product test also covers the source's case $P(A) = 0$ or $P(B) = 0$.

Independent is not the same as mutually exclusive

RETURN TO THE ODD AND EVEN EVENTS

Mutually exclusive

$B = \{1, 3, 5\}$ and $C = \{2, 4, 6\}$

No result can be both odd and even.

$$B \cap C = \emptyset$$

Not independent

$$P(B) = 1/2$$

$$P(B | C) = 0$$

Knowing “even” rules out “odd.”

$$P(B \cap C) = 0 \neq (1/2)(1/2)$$

Two mutually exclusive events with positive probabilities cannot be independent.

The multiplication rule finds a joint probability

“A AND B” MEANS $A \cap B$

$$P(A \cap B) = P(A) P(B | A)$$

First factor: $P(A)$

Probability that A occurs.

×

Second factor: $P(B | A)$

Probability of B within the case A.

$$\text{Also: } P(A \cap B) = P(B) P(A | B)$$

The general rule does not require independence: the conditional factor retains dependence.

Three stages: condition on the full previous history

APPLY THE TWO-EVENT RULE TWICE

$$\begin{aligned} &P(A_1 \cap A_2 \cap A_3) \\ &= P(A_1 \cap A_2) P(A_3 \mid A_1 \cap A_2) \\ &= P(A_1) P(A_2 \mid A_1) P(A_3 \mid A_1 \cap A_2) \end{aligned}$$

Stage 1

No earlier event

Stage 2

Given A_1

Stage 3

Given A_1 and A_2

The last factor is not generally $P(A_3 \mid A_2)$: retain all preceding information.

For independent events, multiplication simplifies

USE UNCONDITIONAL FACTORS ONLY WHEN INDEPENDENCE IS JUSTIFIED

$$P(A \cap B) = P(A) P(B \mid A)$$

Independence gives $P(B \mid A) = P(B)$.

$$P(A \cap B) = P(A) P(B)$$

For a mutually independent collection $A_1, A_2, \dots, A_n$:

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2) \dots P(A_n)$$

For more than two events, pairwise independence alone is not enough.

Example 9: Two resistors selected one by one

SOURCE MODEL: RANDOM DRAWS WITHOUT REPLACEMENT

10 × 50 Ω

5 × 100 Ω

15 resistors total

F_1, F_2 : the first / second resistor is 50 Ω.

H_1, H_2 : the first / second resistor is 100 Ω.

(a) Find $P(H_1)$.

(b) Find $P(H_2 | F_1)$.

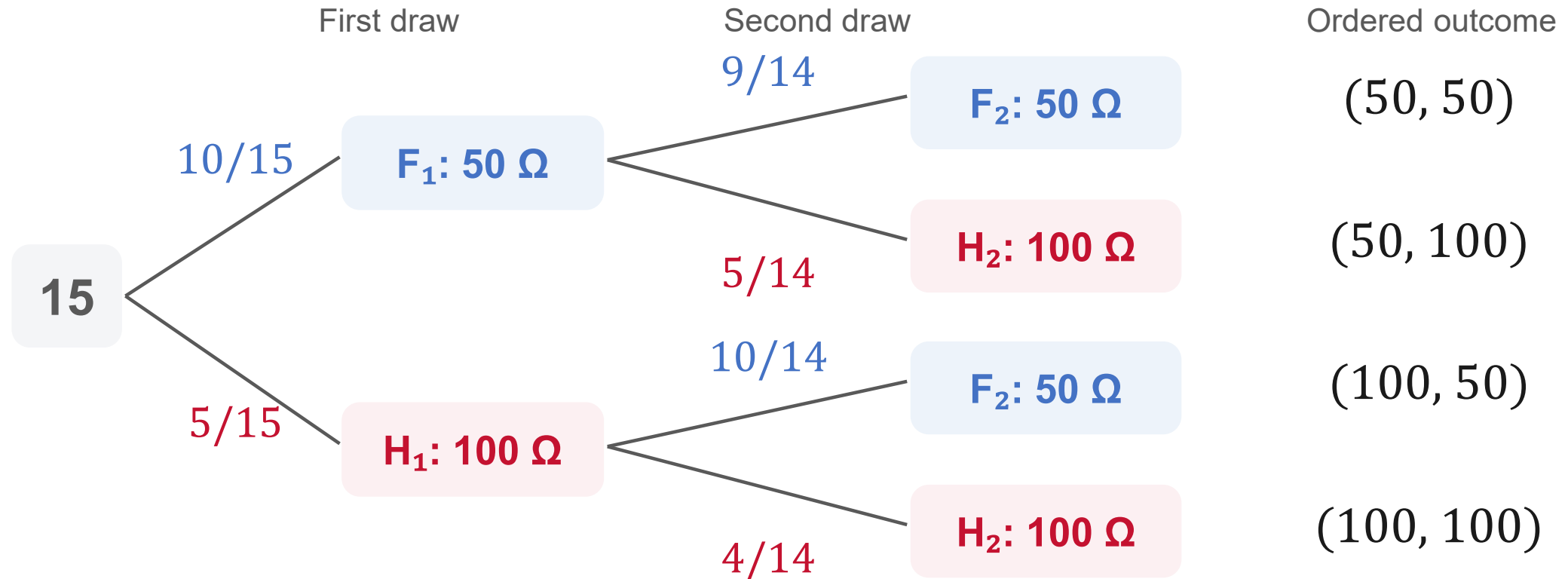
(c) Find $P(H_2 | H_1)$.

(d) Find the probability of each ordered resistance outcome.

(e) Check that the four probabilities sum to 1.

Example 9: Build the conditional-probability tree

FIRST DRAW → UPDATE THE BOX → SECOND DRAW



At every node, the probabilities on the outgoing branches sum to 1.

Example 9: Multiply along paths, then check the total

THE FOUR RESISTANCE-CATEGORY OUTCOMES ARE NOT EQUALLY LIKELY

Outcome	Joint event	Product	Probability
$O_1: (50, 50)$	$F_1 \cap F_2$	$(10/15)(9/14)$	$90/210 = 3/7$
$O_2: (50, 100)$	$F_1 \cap H_2$	$(10/15)(5/14)$	$50/210 = 5/21$
$O_3: (100, 50)$	$H_1 \cap F_2$	$(5/15)(10/14)$	$50/210 = 5/21$
$O_4: (100, 100)$	$H_1 \cap H_2$	$(5/15)(4/14)$	$20/210 = 2/21$

$$(90 + 50 + 50 + 20) / 210 = 1$$

Example 9: Does the first draw affect the second?

USE THE COMPLETED TABLE TO TEST INDEPENDENCE

The event H_2 can occur through two disjoint paths:

$$\begin{aligned} P(H_2) &= P(F_1 \cap H_2) + P(H_1 \cap H_2) \\ &= 50/210 + 20/210 = 1/3 \end{aligned}$$

Given F_1

$$P(H_2 \mid F_1) = 5/14$$

Given H_1

$$P(H_2 \mid H_1) = 4/14$$

Both conditional probabilities differ from $1/3$: the two draws are dependent.

Example 10: Five microprocessors must all work

20% DEFECTIVE • FIVE CHOSEN • FUNCTIONING ASSUMED INDEPENDENT

For each processor: $P(\text{works}) = 1 - 0.20 = 0.80$.

W_1

0.80

×

W_2

0.80

×

W_3

0.80

×

W_4

0.80

×

W_5

0.80

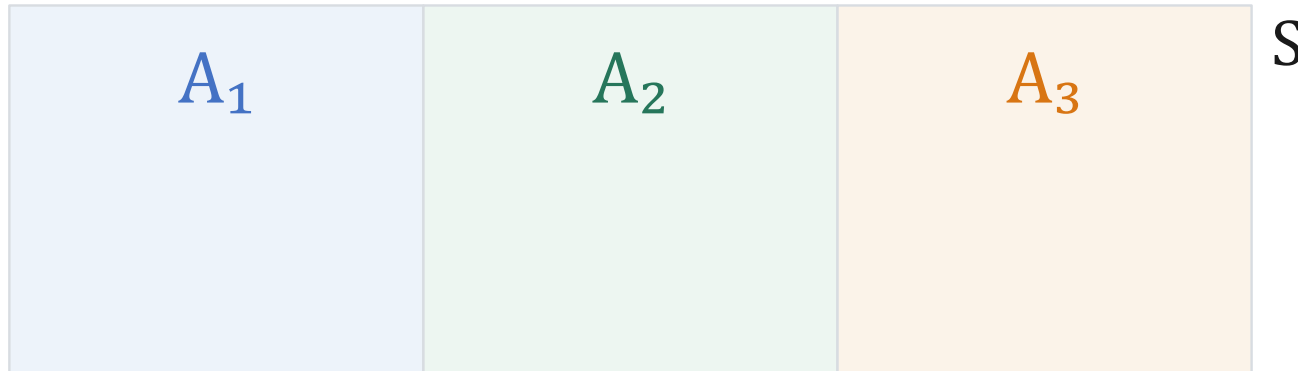
$$P(\text{all five work}) = 0.80^5 = 0.32768$$

About 32.77% under the stated independence model—not 80%.

A partition covers the sample space without overlap

MUTUALLY EXCLUSIVE AND EXHAUSTIVE EVENTS

Each outcome belongs to exactly one event in the partition.



$$A_i \cap A_j = \emptyset \text{ when } i \neq j$$

$$A_1 \cup A_2 \cup \cdots \cup A_n = S$$



A and A^c also form a partition.

Coffee example: Cup size defines a partition

SOURCE EXAMPLE: ONE RANDOMLY SELECTED CUP

Every cup is small, medium, or large; no cup has two sizes.

A_1 : Small

30%

$$P(A_1) = 0.30$$

A_2 : Medium

35%

$$P(A_2) = 0.35$$

A_3 : Large

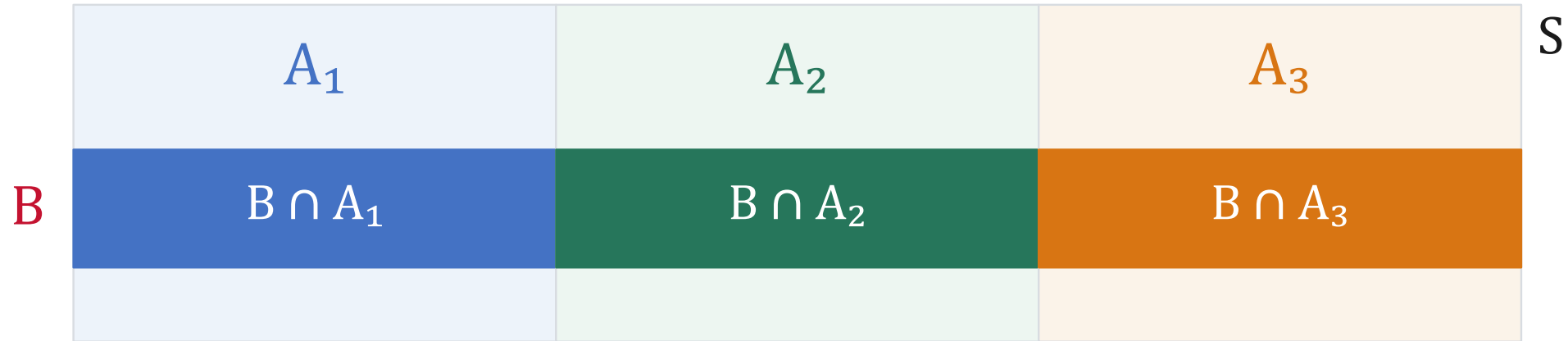
35%

$$P(A_3) = 0.35$$

Yes, these events form a partition: $0.30 + 0.35 + 0.35 = 1$.

The law of total probability adds disjoint contributions

BREAK B INTO ITS PARTS INSIDE $A_1, A_2, \dots, A_n$



$$\begin{aligned} P(B) &= P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3) \\ &= P(A_1)P(B | A_1) + P(A_2)P(B | A_2) + P(A_3)P(B | A_3) \end{aligned}$$

In general: $P(B) = \sum_i P(A_i)P(B | A_i)$, where $i = 1, \dots, n$.

Coffee example: What is the probability of regular coffee?

B = REGULAR COFFEE; B^c = DECAF

Size	Size probability	Regular-coffee probability within size
Small, A_1	0.30	$P(B \mid A_1)$
Medium, A_2	0.35	$P(B \mid A_2)$
Large, A_3	0.35	$P(B \mid A_3)$

$$P(B) = 0.30P(B \mid A_1) + 0.35P(B \mid A_2) + 0.35P(B \mid A_3)$$

A numerical result requires the three within-size coffee preferences.

Bayes' rule reverses the conditioning

FROM $P(B | A_k)$ TO $P(A_k | B)$

$$P(A_k | B) = \frac{P(A_k) P(B | A_k)}{P(B)}$$

Use total probability for the denominator:

$$P(A_k | B) = \frac{P(A_k) P(B | A_k)}{\sum_j P(A_j) P(B | A_j)}$$

$j = 1, \dots, n$; $P(B) > 0$; $A_1, \dots, A_n$ form a partition.

Coffee example: Among regular orders, which cup size?

THE GIVEN EVENT IS NOW B = REGULAR COFFEE

The reference group is regular-coffee orders—not all orders.

Requested probability	Joint contribution divided by $P(B)$
$P(A_1 B)$: small	$0.30P(B A_1) / P(B)$
$P(A_2 B)$: medium	$0.35P(B A_2) / P(B)$
$P(A_3 B)$: large	$0.35P(B A_3) / P(B)$

The three conditional size probabilities sum to 1 when $P(B) > 0$.

Two directions through the same probability model

SAME JOINT PROBABILITIES • DIFFERENT QUESTIONS

Total probability

Given the group shares and within-group probabilities:

What is the overall $P(B)$?

Add all joint contributions.

Bayes' rule

Given that B occurred:

Which group did the outcome belong to?

Divide one contribution by their total.

Example 11: Emissions-test failures across engine sizes

CLASSROOM MODEL RETAINED FROM THE SOURCE

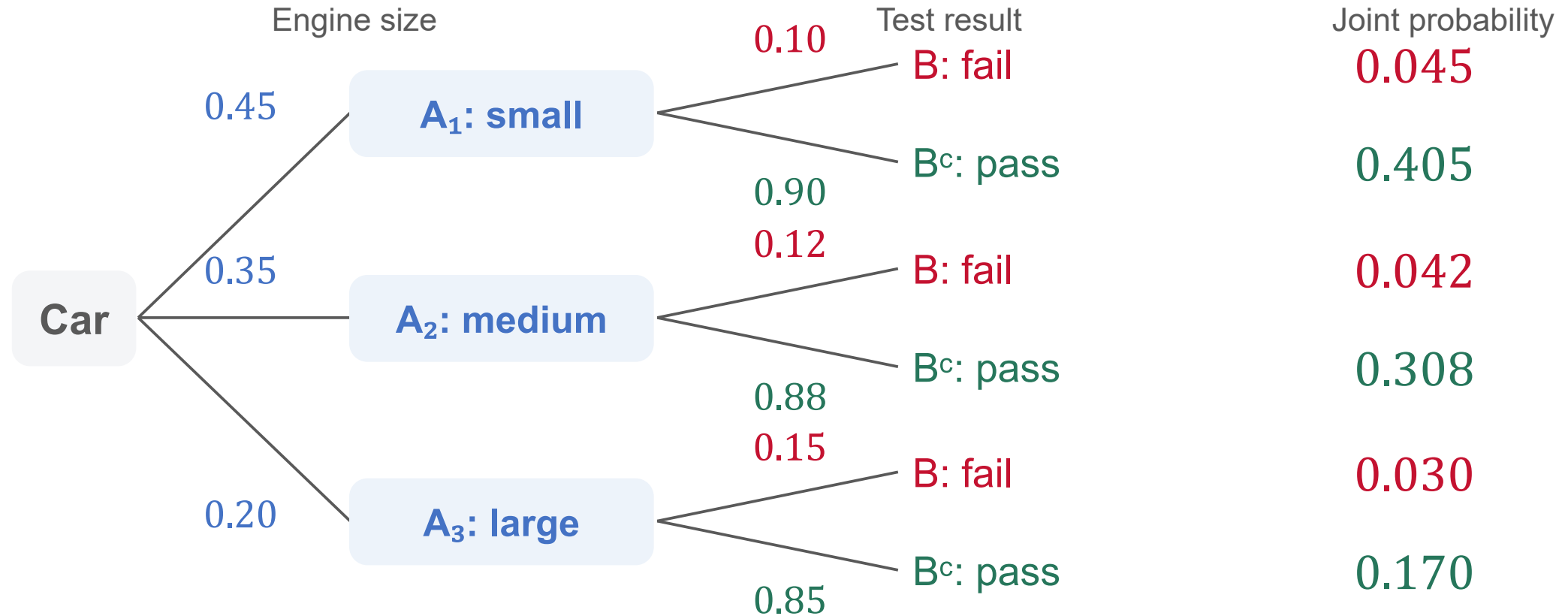
A car of this make has one of three engine sizes. Let B be failure of an emissions test within two years of purchase.

Engine size	Event	Share of cars	Failure rate within size
Smallest	A_1	0.45	$P(B A_1) = 0.10$
Medium	A_2	0.35	$P(B A_2) = 0.12$
Largest	A_3	0.20	$P(B A_3) = 0.15$

Question: What is $P(B)$ for a randomly selected car of this make?

Example 11: The tree separates size and test result

MULTIPLY ALONG EACH COMPLETE PATH



Example 11: Calculate the overall failure probability

WEIGHT EACH FAILURE RATE BY ITS ENGINE-SIZE SHARE

$$P(B) = (0.45)(0.10) + (0.35)(0.12) + (0.20)(0.15)$$

Small									
0.045		+		0.042		+		0.030	

$$P(B) = 0.117 = 11.7\%$$

The complementary pass probability is $P(B^c) = 1 - 0.117 = 0.883$.

Example 12: Reverse the question after seeing the result

USE THE SAME ENGINE-SIZE AND EMISSIONS MODEL

(a) A failed-test record

What is the probability that the car has the smallest engine?

Find $P(A_1 | B)$.

(b) A passed-test record

What is the probability that the car has the smallest engine?

Find $P(A_1 | B^c)$.

Before calculating, name the reference group—and therefore the denominator.

Example 12: Two different conditioning groups

THE NUMERATOR MUST MATCH THE OBSERVED TEST RESULT

Failed record: $P(A_1 | B)$

$$\frac{(0.45)(0.10)}{0.117}$$

$$0.3846 \approx 38.46\%$$

Passed record: $P(A_1 | B^c)$

$$\frac{(0.45)(0.90)}{0.883}$$

$$0.4587 \approx 45.87\%$$

$P(A_1 | B^c)$ is not $1 - P(A_1 | B)$: the conditioning group has changed.

Example 12: Read the denominator directly from a table

CELLS ARE JOINT PROBABILITIES, NOT WITHIN-ROW PERCENTAGES

Engine size	Fail B	Pass B ^c	All cars
Smallest, A ₁	0.045	0.405	0.450
Medium, A ₂	0.042	0.308	0.350
Largest, A ₃	0.030	0.170	0.200
Total	0.117	0.883	1.000

Among failures: 0.045 / 0.117

Among passes: 0.405 / 0.883

Example 13: A positive test—what does it imply?

HYPOTHETICAL CLASSROOM MODEL • NOT CLINICAL ADVICE

D: the person has the disease.

T⁺: the test produces a positive signal.

Disease proportion

$$P(D) = 0.005$$

0.5% of the community

Positive when diseased

$$P(T^+ | D) = 0.99$$

99% within D

Positive when not diseased

$$P(T^+ | D^c) = 0.01$$

1% within D^c

Question: If the test is positive, what is $P(D | T^+)$?

Example 13: Count both routes to a positive result

FIRST FIND THE TOTAL POSITIVE-TEST PROBABILITY

Route to T^+	Joint probability
Disease and positive	$(0.005)(0.99) = 0.00495$
No disease and positive	$(0.995)(0.01) = 0.00995$
All positive results	$P(T^+) = 0.01490$

$$P(D | T^+) = \frac{0.00495}{0.01490} \approx 0.3322 = 33.22\%$$

99% answers the forward question; about 33.22% answers the reverse question.

Example 13: Make the calculation tangible with counts

ILLUSTRATIVE EXPECTED COUNTS FOR 20,000 TESTED PEOPLE

Apply the source probabilities to a hypothetical group of 20,000.

Group	Positive	Negative	Total
Disease D	99	1	100
No disease D ^c	199	19,701	19,900
Total	298	19,702	20,000

Among the 298 positives, 99 have the disease: $99/298 \approx 33.22\%$.

Choose the rule by reading the question

DEFINE THE EVENT AND ITS REFERENCE GROUP FIRST

Question asks for...	Use...
A given B	$P(A B) = P(A \cap B) / P(B)$
A and B	$P(A \cap B) = P(A)P(B A)$
All independent events	Product of their individual probabilities
B across a partition	$P(B) = \sum_i P(A_i)P(B A_i)$
Group A_k , given B	$P(A_k B) = P(A_k)P(B A_k) / P(B)$

“Given” changes the denominator. “And” identifies a joint event.

Concept check: Diagnose the error before calculating

EACH STATEMENT BELOW IS INCORRECT

A. Same fair die

$P(A | B)$ must equal $P(B | A)$.

B. Two 100 Ω resistors

Without replacement, use $(5/15)^2$.

C. Overall emissions failure

Add $0.10 + 0.12 + 0.15$.

D. Positive diagnostic test

$P(D | T^+)$ is 0.99.

Concept check: Correct rule and reasoning

EXPLAIN THE REFERENCE GROUP OR ASSUMPTION

A $P(A | B) = 1/3; P(B | A) = 1$

Changing the condition changes the reference set.

B $(5/15)(4/14) = 2/21$

After selecting 100 Ω first, four remain among fourteen.

C $(0.45)(0.10) + (0.35)(0.12) + (0.20)(0.15) = 0.117$

Use total probability, with engine-size shares as weights.

D $0.00495 / 0.01490 \approx 0.3322$

Include both routes to a positive result in the denominator.

Lecture 6: Key takeaways

ONE MODEL, SEVERAL WAYS TO ASK ABOUT IT

Condition

$P(A | B)$ uses B as the reference group.

Independence

Knowing another event does not change the probability.

Multiply

Along a path, retain the appropriate conditioning information.

Total probability

Add disjoint joint contributions across a partition.

Bayes' rule

Divide the selected joint contribution by the observed-event total.

Always check: the condition, the assumptions, and whether the result lies in $[0, 1]$.